

AYE
TECH HUB

• AI & PRODUCTIVITY • LESSON 08

AI Image Generation

Midjourney · DALL-E 3 · Stable Diffusion · Adobe Firefly

Lesson 08 of 30 | AI & Productivity Program

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- How diffusion models turn text into photorealistic images
- Platform deep-dive: Midjourney, DALL-E 3, Stable Diffusion, Firefly, Ideogram
- Prompt engineering for images — style, lighting, camera, and mood keywords
- Negative prompts, aspect ratios, seed values, and style references
- Commercial licensing, copyright, ethics, and responsible AI image use

What is AI Image Generation?

AI image generation is the technology that creates visual images — photographs, illustrations, logos, artwork — from text descriptions. You type what you want to see, and a model produces it in seconds. This technology has advanced from blurry, distorted faces in 2021 to photorealistic 8K images in 2024-2026. **It is not magic — it is mathematics.** The model learns by seeing hundreds of millions of image-text pairs, then learns to reverse a noise process to generate new images.

Stable Diffusion / DALL-E 3 / Midjourney all use variations of this denoising diffusion process

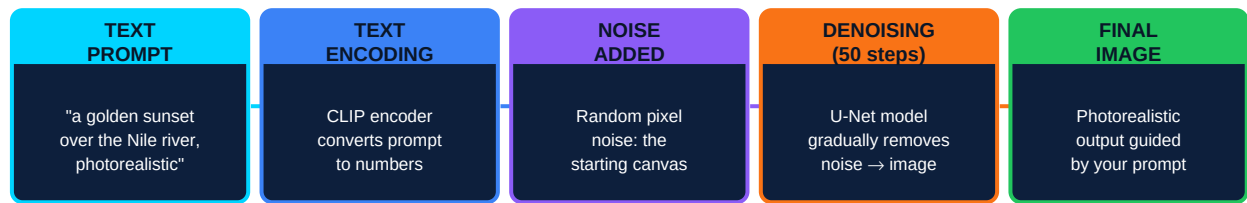


Figure 1 — How Diffusion Models Generate Images from Text (Simplified)

Term	What It Means	Practical Impact
Diffusion Model	AI architecture that learns to remove random noise from images step-by-step, guided by a text description. Used by Stable Diffusion, DALL-E 3, Midjourney.	The same base technology powers almost all modern AI image tools — understanding it helps you use all platforms better
Latent Space	The compressed mathematical "space" where the model stores its understanding of what images look like. All generation happens here before converting to pixels.	Editing in latent space (img2img) lets you transform existing images — change style, add objects, alter composition
CLIP (text encoder)	The model component that converts your text prompt into numbers the diffusion model understands. Determines how well prompt → image alignment works.	Better CLIP → better prompt following. DALL-E 3 and Ideogram have the best prompt adherence; Midjourney interprets more creatively
Steps / Iterations	The number of denoising passes the model takes. More steps = more detail but slower. Most models need 20-50 steps for good quality.	For quick previews use 20 steps. For final images use 40-50. Beyond 60 steps rarely improves quality
CFG Scale / Guidance	How strictly the model follows your prompt. Low CFG (1-4) = creative, loose. High CFG (10-20) = literal but can over-saturate.	Start at 7 for balanced results. Increase to 10-12 for more literal prompt following. Decrease for artistic, dreamlike output
Seed Value	A number that sets the initial random noise pattern. Same seed + same prompt = same image. Different seed = different variation.	Save your seed when you get a result you like — you can then vary only the prompt while keeping the same composition

The most important insight: AI image models do not "understand" images the way humans do. They learn statistical patterns between text and pixels. This is why prompts matter enormously — the model cannot read your mind, it can only respond to the words and patterns it was trained on.

Platform Deep-Dive and Comparison

Five platforms dominate the AI image generation space in 2026. Each has a different strength, pricing model, and ideal use case. Professionals often use more than one depending on the task.

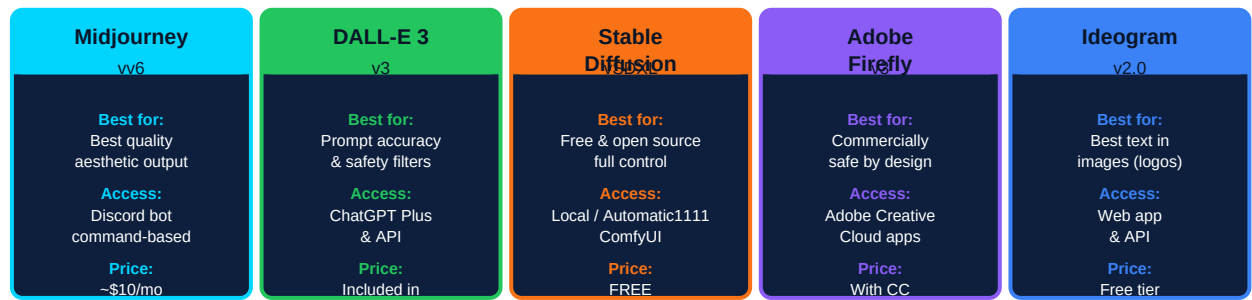


Figure 2 — AI Image Generation Platform Comparison (2026)

Platform	Best Use Case	Weaknesses	Who Should Use It
Midjourney v6	Artistic photography, concept art, marketing visuals, landscapes. Best for aesthetic quality and cinematic imagery.	No free tier. Discord-only interface (v6). Less accurate prompt following than DALL-E 3. Opaque model.	Designers, content creators, social media managers, anyone needing consistently beautiful output
DALL-E 3 (via ChatGPT)	Infographics, text in images, product visualisation, accurate scene composition following complex prompts.	Images have a recognisable "clean" style. Heavy safety filters. Cannot do violence, nudity, or celebrities.	Marketers, educators, office professionals, anyone already using ChatGPT Plus
Stable Diffusion (local)	Full creative control, NSFW, custom LoRA models, batch generation, fine-tuning on custom data.	Requires technical setup and a good GPU (6GB+ VRAM). Quality varies by model. Steep learning curve.	Developers, researchers, power users, anyone needing full control or privacy
Adobe Firefly	Commercially safe illustrations for professional work — ad campaigns, stock image alternatives, Adobe workflow.	Less photorealistic than Midjourney. Style is more "stock photo" than artistic.	Designers using Adobe CC, agencies needing images with clean commercial licensing
Ideogram 2.0	Logos, posters, social graphics — anything requiring readable text inside an image (other tools fail at this).	Not as strong as Midjourney for photorealistic photography. Smaller community and model library.	Brand designers, logo creators, social media graphics with text overlay

Midjourney Commands	What They Do
/imagine prompt:	Generate 4 image variations from your prompt
--ar 16:9 / --ar 9:16 / --ar 1:1	Set aspect ratio (widescreen / portrait / square)
--style raw	Less Midjourney "aesthetic filter" — more literal prompt
--chaos 0-100	Higher = more varied and unexpected compositions
--seed 12345	Lock the random seed for reproducible results
--no hands, blurry	Negative prompts — tell model what to exclude
--sref [image URL]	Style reference — match the style of another image
U1 U2 U3 U4 / V1 V2	Upscale / create 4 variations of a specific grid image

Prompt Engineering for Images

The quality of your output depends almost entirely on the quality of your prompt. A weak prompt gives generic, disappointing images. A structured prompt with subject, style, lighting, camera, and quality keywords consistently produces professional-grade results.

Figure 3 — Anatomy of a Well-Structured AI Image Prompt



Negative Prompts — What NOT to Include

Category	Common Negative Prompt Terms
Quality issues	blurry, out of focus, low quality, low resolution, pixelated, JPEG artifacts, overexposed, underexposed, washed out, grainy
Anatomy errors	extra fingers, deformed hands, missing limbs, distorted face, asymmetrical eyes, extra limbs, bad anatomy, disfigured
Style conflicts	cartoon (if you want realistic), sketch, rough, unfinished, watermark, logo, text, signature, username, frame, border
Generation artifacts	duplicate, multiple subjects (when you want one), crowded, busy background (when you want clean), dark, underlit (when you want bright)

Ethics & Copyright Topic	Current Reality & Best Practice
Copyright of AI images	In most countries (including USA as of 2026) AI-generated images have no automatic copyright unless there is substantial human creative input. You can still use them commercially — but cannot sue if others copy them.

Training data lawsuits	Major lawsuits against Stability AI, Midjourney, and others for training on copyrighted images without permission. Outcome still evolving. Adobe Firefly is the safest — trained only on licensed stock images.
Deepfakes and misuse	Generating realistic images of real people without consent is illegal in many countries and violates platform terms. All major platforms prohibit realistic non-consensual imagery of named individuals.
Disclosure best practice	In professional contexts, disclose when images are AI-generated. Mislabelling AI images as photographs in journalism or advertising is considered unethical and may be illegal under consumer protection laws.
Recommended approach	Use Adobe Firefly or Midjourney (commercially licensed) for client work. Use Stable Diffusion locally for experimentation. Always disclose in professional contexts. Avoid generating images of real, identifiable people.

What comes next — AI-009: AI Video Generation — Runway ML, Kling AI, Sora, Pika Labs, and HeyGen. How to generate consistent AI video clips, animate still images, create AI avatar videos, and build short-form video content entirely with AI tools.